AndMTProto · UI proposal A

Evolution: one clear screen

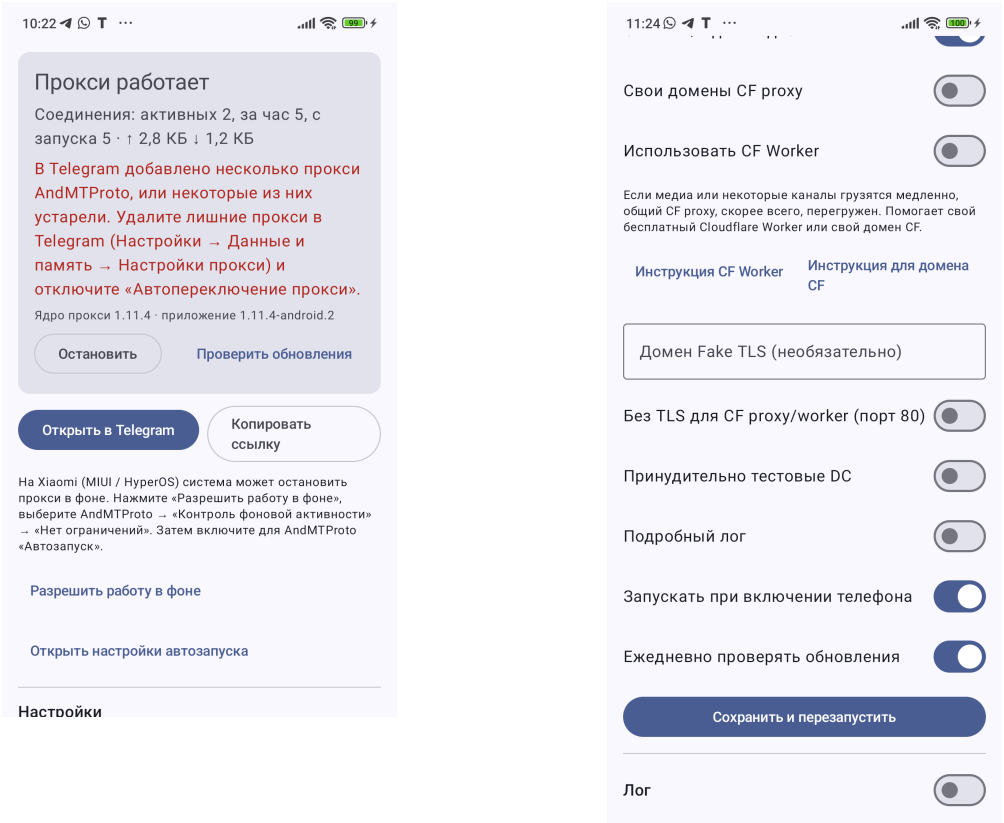
Keep the single-screen app, rebuild it around a big status and connect control, and move settings to their own grouped screen.

Scope	Main screen redesign, separate Settings screen, log as a bottom sheet, attention cards instead of hint text.
Effort	About 1–2 weeks for one developer; no Python changes.
Risk	Low. Same navigation model, same data, incremental release possible.
Best for	Shipping a visibly modern app quickly without re-architecting.

Prepared 7 November 2026 for the AndMTProto maintainer. Baseline: AndMTProto 1.11.4-android.3 (main, with the daily update check from PR #19). Screens in this document are wireframe mockups; labels are in Russian because that is the app's primary language.

1. Where the UI stands today

The whole app is one long scrolling screen built with Jetpack Compose and Material 3 (dynamic colours on Android 12+). From top to bottom: the status card, the Telegram buttons, a block of background-work hints, about twenty settings, the Save button and the log. It works, and every recent feature found a place on it, but it has grown by accretion. The screenshots below are from the test phone (Redmi Note 10 Pro, MIUI 14) on 7 November 2026.



Left: status card with the stale-proxy hint. Right: the lower half of the settings. (The secret field is cropped out.)

Problems worth fixing

Problem	Effect on users
No single glanceable state. The status is a text title; Start/Stop is a small button next to “Проверить обновления”.	Users can't tell at a glance whether Telegram is protected; the most important action has the same weight as a secondary one.
Settings and status share one screen. Twenty-plus fields, most of them advanced (pool size, buffer, test DCs, no-TLS).	The everyday screen is long and intimidating; mistakes in advanced fields are easy (e.g. a non-loopback address).
The secret is shown in plain text on the main screen.	Any screenshot of the main screen shared for support leaks it, and with it access to the proxy.
Hints are walls of text (MIUI background work, CF overload, stale proxies).	They push the controls down, stay after the problem is solved and get ignored.
Route health is invisible. Whether DC2 goes over WebSocket or Cloudflare, or the shared CF proxy fails, is only in the log.	When Telegram is slow, users can't see why; support needs log screenshots.

Problem	Effect on users
Setup is spread out. Start → open in Telegram → battery → MIUI autostart are separate buttons with no sense of progress.	First-run users miss the background-work steps, then the proxy dies in the background.
Layout shifts after Save and after Stop (seen while testing).	Mis-taps: one landed on “Открыть в Telegram”.

2. What modern Android UI looks like in 2026

Platform and Material guidance

- **Material 3 Expressive** extends Material 3 with spring-based motion, shape morphing, emphasised type and new components. Google's guidance is Compose-first. Caveat: much of the expressive API was still alpha or experimental in Compose Material3 in 2026, so a proposal should use it only where it degrades gracefully.
- **Edge-to-edge is mandatory** for apps targeting Android 16 (AndMTProto targets API 36), and **predictive back** animations are on by default. The UI must handle insets and back gestures properly; a multi-screen design should use the navigation components that support predictive back.
- **Progress-centric notifications** (Notification.ProgressStyle, the basis of Live Updates) exist since Android 16. They aren't a natural fit for an always-on proxy, but the foreground notification can carry richer state (route, traffic).
- **Dynamic colour** stays the norm on Android 12+, with a brand palette as fallback; AndMTProto already does this.

Patterns from VPN and proxy apps

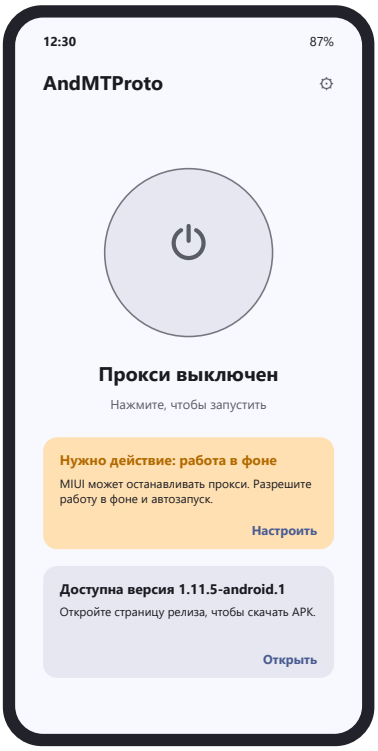
- One unmistakable **connect control** in the centre, with a clear connected / disconnected state and simple stats beside it.
- **Explicit transitional states** ("Подключение…") with animation, never a status that misrepresents the real connection. AndMTProto already distinguishes "starting" from "listening"; the UI should show it.
- Live **timer and traffic** readouts reassure users that it works.
- Redesigns (e.g. NordVPN 2025) concentrated on the connection screen and first-time use rather than on settings.

Constraints specific to AndMTProto

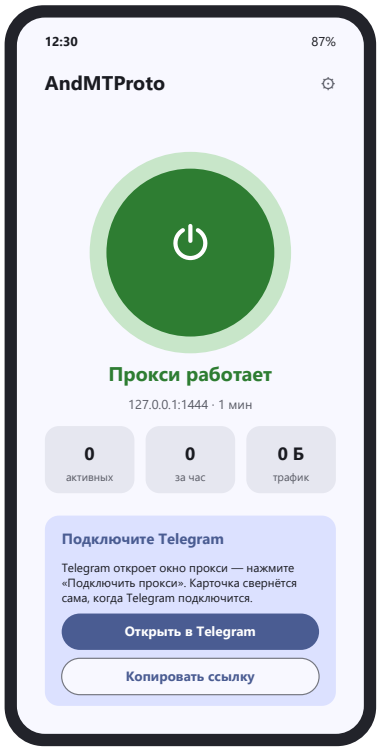
- Users are often non-technical and on Xiaomi phones, where background restrictions are the main failure; the UI has to guide them through MIUI settings.
- Russian first, English second; Russian strings are 20–30 % longer, so layouts need room.
- Play Protect passed review; the redesign must not add permissions, trackers or network calls.
- The Python core is upstream code: new UI data must come from the adapter (android_entry.py), never from patching proxy/.

3. The proposal

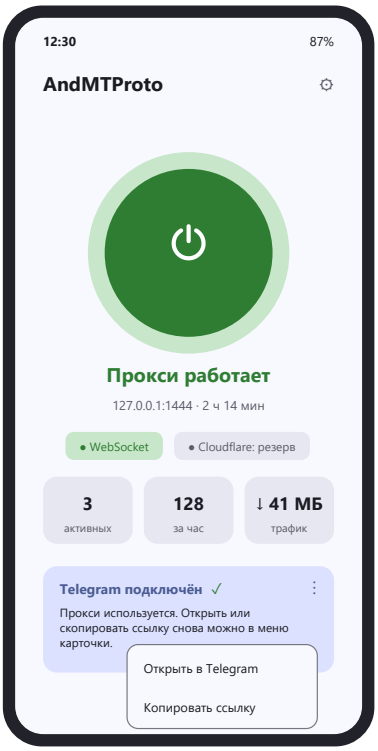
Proposal A keeps everything a user does every day on one screen and moves everything they set once into Settings. The main screen becomes three layers: **state** (a big connect control with a status line), **health** (route chips and three stat tiles) and **attention** (cards that appear only when something needs doing).



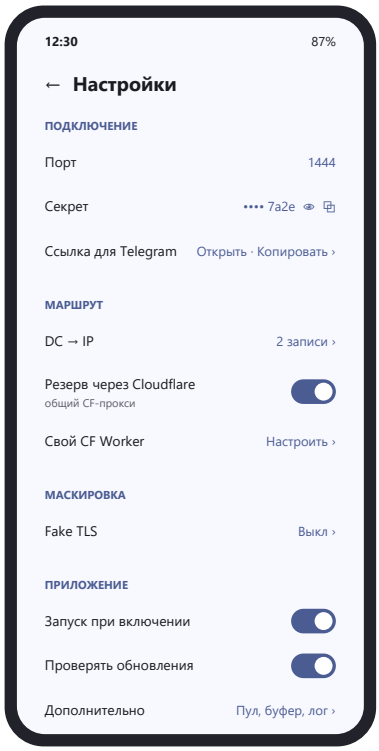
1 Off, with attention cards



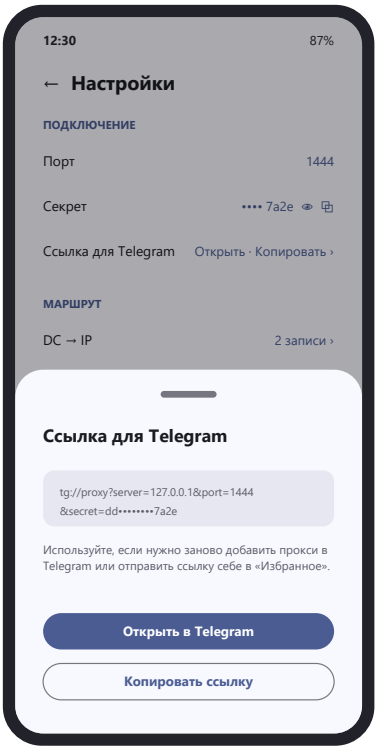
2 Running, connect Telegram



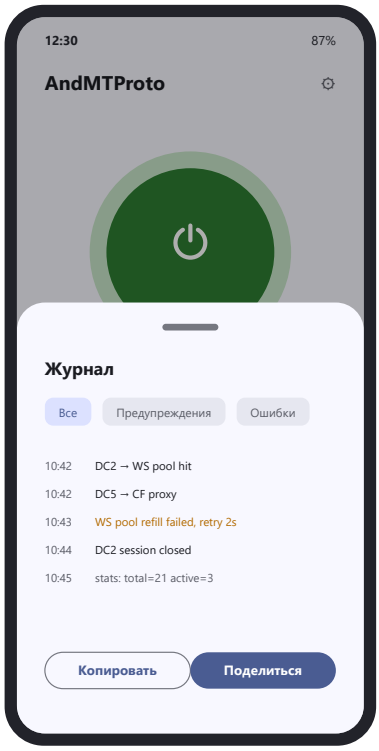
3 Running, Telegram connected



4 Settings, grouped



5 Telegram link from Settings



6 Log as a bottom sheet

Screen by screen

Screen	What changes
1 Off	A 96 dp connect control replaces the small Start button. Attention cards (background work, update available, errors with a fix button) stack below and disappear once resolved.
2 Connect Telegram	Right after the proxy starts, until Telegram connects: a “Подключите Telegram” card with both actions as full-width buttons, **Открыть в Telegram** (filled) and **Копировать ссылку** (outlined).
3 Running	The control turns green with a halo; the status line shows address and uptime. Route chips show WebSocket / Cloudflare health from the existing counters. Stat tiles: active, last hour, traffic. Once Telegram connects, the card collapses to “Telegram подключён” with a tick; both actions stay in its three-dot menu.
4 Settings	Its own screen (gear icon), grouped: Подключение, Маршрут, Маскировка, Приложение, Дополнительно. The secret is masked with show and copy buttons. Rare options (pool, buffer, test DC, no-TLS, verbose log) move under “Дополнительно”. Changes save on leave with a “перезапустить?” snackbar instead of a big Save button.
5 Link sheet	Settings → Подключение → “Ссылка для Telegram” opens a sheet with the link (secret masked) and the same two buttons. This is the permanent place for both actions, whatever the main screen shows.
6 Log	A bottom sheet with filter chips and Copy / Share; the secret is already redacted in the log (since 1.11.4-android.2).

Where “Открыть в Telegram” and “Копировать ссылку” live

- **Before Telegram connects** (screen 2): both as big buttons on the main screen, because adding the proxy to Telegram is the one thing left to do.
- **After Telegram connects** (screen 3): the card collapses; both actions are in its three-dot menu. “Connected” means at least one connection since start that wasn't rejected (the existing counters: total minus rejected handshakes).
- **Always** (screen 5): Settings → Подключение → “Ссылка для Telegram”.

- The collapsed state is remembered across proxy restarts and resets when the port or secret changes, because Telegram then needs the new link. The stale-proxy hint (PR #18) also expands the card again, since re-adding the proxy is its fix.
- The notification keeps its "Открыть в Telegram" action as today.

4. Interaction and visual details

- **States:** stopped (neutral circle) → starting (rounded square with a spinning arc, "Запуск…", mirrors the adapter's listening flag) → running (green circle) → error (red, with the error card). The shape change uses Material 3 Expressive shape morphing where available and a plain crossfade otherwise.
- **Stable layout:** the connect control and status line have fixed height; cards animate in below, so taps never shift onto other buttons.
- **Colour:** dynamic colour on Android 12+, the current indigo palette elsewhere; semantic green / amber / red only for state, never for decoration.
- **Accessibility:** the control announces its state ("Прокси работает, кнопка, остановить"); every touch target ≥ 48 dp; text scales to 200 % without clipping (Russian strings are long).
- **Edge-to-edge and predictive back:** Settings uses a standard Compose navigation destination, so the system back preview works.
- **Notification:** unchanged layout, plus the route in the text ("WebSocket · 3 активных").

5. Implementation plan

Step	Work	Days
1	Navigation: add androidx.navigation-compose (or a two-state host) with Main and Settings destinations; predictive back.	1
2	Main screen: ConnectControl composable with four states, status line, route chips, stat tiles; fixed-height header.	2–3
3	Attention cards: one AttentionCard component; move MIUI, battery, CF overload, stale proxy, non-loopback and update hints into it with dismiss rules. Telegram link card with its expanded / collapsed rule; link sheet in Settings.	2
4	Settings screen: regroup SettingsForm, secret masking, "Дополнительно" subpage, save-on-leave with restart snackbar.	2
5	Log bottom sheet with filters (parse level from each line).	1
6	Strings RU/EN, accessibility pass, phone test on MIUI, release.	1–2

No changes to the Python adapter or upstream code. Route chips use the counters status() already returns (ws, cf, pool); if needed, the adapter can add them to status() in a few lines. One new dependency at most (navigation-compose).

Risks

- Material 3 Expressive APIs may still be experimental in the Compose BOM in use; the design works with stable Material 3 alone.
- Moving settings off the main screen changes habits; the first launch after the update can show a one-time "Настройки переехали" tip.

6. How it compares with proposal B

	A · Evolution	B · Dashboard
Navigation	One screen + Settings	Bottom bar, four tabs
First-run guidance	Attention cards	Step-by-step wizard
Route visibility	Two chips	Per-DC table, latency, self-test
Python adapter changes	None (optional status fields)	New per-DC status and self-test
Effort	1–2 weeks	3–4 weeks
Risk	Low	Medium

Recommendation: A first. It fixes the problems users actually hit (unclear state, long screen, visible secret, missed background setup) in a release or two. B's network view and self-test can follow later as an extra screen on top of A.

Sources

Android Developers Blog — Android 16 is here (edge-to-edge, predictive back, ProgressStyle)
https://android-developers.googleblog.com/2025/06/android-16-is-here.html

Material 3 skill wiki (community summary of Material 3 Expressive / I/O 2026 guidance)
https://github.com/hamen/material-3-skill/wiki

Material Design 3 overview (third-party summary)
https://aksite.pages.dev/ai-native-design/material-design/

VPN connection switch UI notes (VP0; promotional, used as a starting point)
https://vp0.com/blogs/vpn-connection-switch-ui-clone-ios

HeroVPN app design case study
https://contra.com/p/gvr23BQg-hero-vpn-secure-vpn-app-design

VPNIFY UX teardown
https://screensdesign.com/showcase/vpnify-unlimited-vpn

Bitmask VPN visual design proposal
https://0xacab.org/leap/bitmask-vpn/-/issues/513

Tom's Guide — NordVPN mobile app redesign
https://www.tomsguide.com/computing/vpns/nordvpn-announces-mobile-app-redesign-heres-what-you-need-to-know

Material 3 Expressive details come from secondary summaries; check the AndroidX Compose Material3 release notes for the exact API status before implementation.